

School of Mathematics and Statistics, UNSW
MATH3311/MATH5335 Python
Term 2, 2023

Exercises 3: Matrix structure and matrix factorizations

1. Download the binary file `corr.mat` and `mat2numpy.py` (this is created from the `corr.mat` file) from the course web site and save them in the same folder.

In this question, you are asked to perform a number of tests and then give your opinion, with reasons, as to whether the matrix is a valid correlation matrix. Start by creating `.py` file (`ex03q01.py`) where the following libraries might be needed.

`from numpy import maximum, finfo, all, flip, arange, count_nonzero, min, max, floor, log10, prod, diag, eye`

`from numpy.linalg import cond, eig`

`from scipy.linalg import cholesky, det, inv, LinAlgError`

`from matplotlib.pyplot import semilogy, title, grid`

`from mat2numpy import K`

- (a) What are the dimensions of K ? Is K square?
- (b) Is K symmetric?
- (c) Use the Cholesky factorization $K = R^T R$ to determine if K is positive definite.
- (d) Calculate and plot the eigenvalues of K in **descending** order. Then, find the smallest and largest eigenvalues.
- (e) Calculate the determinant $\det K$ by
 - i. Using the function `scipy.linalg.det`.
 - ii. Using the Cholesky factorization.
 - iii. Using the eigenvalues.Do all methods give you the same answer?
- (f) Are all the diagonal entries of K equal to 1?
- (g) For **different** assets find
 - i. the minimum correlation;
 - ii. the maximum correlation.
- (h) What percentage of the elements are non-zero?
- (i) An application (supposedly) requires the inverse K^{-1} . Calculate K^{-1} using the function `scipy.linalg.inv`. Is K nonsingular? Does the inverse exist?
- (j) The application actually requires $\alpha = \mathbf{b}^T K^{-1} \mathbf{b}$.
 - i. Can this be done without using the inverse?
 - ii. What is the most efficient way to calculate α ?
- (k) Calculate the condition number $\kappa_2(K)$ using the 2-norm?

- i. If all data is known “exactly”, that is, to full double precision accuracy, how many significant figures are there in the solution to $K\mathbf{x} = \mathbf{b}$?
 - (l) Are there any other tests you could apply?
 - (m) Is K a realistic correlation matrix?
2. Download the Python function `laplace0.py` from the course web site. In the parts below, you need to import:

```
from laplace0 import laplace0
from scipy.linalg import cholesky, norm, qr, eigvalsh, LinAlgError, svd, inv, lu
from scipy.sparse import eye
from numpy import prod
from matplotlib.pyplot import subplot, spy, figure, title, semilogy, grid
```

- (a) What does this function do and what input arguments does it require?
 - (b) Create a matrix A with `laplace0.py` using the default arguments.
 - (c) Say as much as you can about the structure of A :
 - i. What are the dimension of A ?
 - ii. Is A stored as a sparse matrix? Create a spy plot of A .
 - iii. Is A symmetric? positive definite? orthogonal?
 - iv. Is A banded? What are the lower, upper and total bandwidth of A ?
 - v. Is A Toeplitz?
 - vi. What are the eigenvalues of A ? Plot them. Repeat for singular values?
 - vii. What is the determinant of A ?
 - viii. Does the inverse of A exist? What does a spy plot of the inverse tell you?
 - ix. Calculate the LU , Cholesky, QR , Spectral and SVD factorizations of A .
3. (a) Write a Python function `matstruc(A, tol)` (in a file script `matstruc.py`) that returns a named tuple, say `s`, with information about the structure of the matrix A . The fields of `s` should be as follows, using `tol` to define the tolerance used when deciding if a floating point quantity is sufficiently close to zero. You may start as:

```
from collections import namedtuple
from numpy import eye, count_nonzero, max, min
from scipy.linalg import norm, cholesky, toeplitz, tril, triu, LinAlgError, eigvals, svdvals
from scipy.sparse import csr_matrix, issparse
```

```
s = namedtuple('s', 'dim1, dim2, square, maxval, minval, sym, posdef, orthog,
bndup, bndlo, trilo, triup, toeplitz, evmin, evmax, svmin, svmax, cond,
sparse, nnz, sparsity')
```

```
def matstruc(A, tol):
```

- `s.dim1` and `s.dim2` gives the first and second dimensions of A , respectively.
- `s.square` is `True` if and only if A is a square matrix.

- `s.maxval` (`s.minval`) are the maximum (minimum) magnitude of the entries of A .
- `s.sym` is `True` if the matrix A is symmetric.
- `s.posdef` is `True` if A is positive definite.
- `s.orthog` is `True` if A is orthogonal.
- `s.bndup` and `s.dndlo` are the upper and lower bandwidths of A .
- `s.trilo` is `True` if A is lower triangular.
- `s.triup` is `True` if A is upper triangular.
- `s.toeplitz` is `True` if A is Toeplitz.
- `s.evmin` and `s.evmax` give the smallest and largest eigenvalues in magnitude.
- `s.svmin` and `s.svmax` give the smallest and largest singular values.
- `s.cond` gives the condition number with respect to the 2-norm.

Test your function with the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 & 0; & -1 & 2 & -1 & 0; & 0 & -1 & 2 & -1; & 0 & 0 & -1 & 2 \end{bmatrix}.$$

(b) Modify your `matstruc` function so that it works when A is stored as a sparse matrix, and add the following fields.

- `s.sparse` is `True` if A is stored as a sparse matrix. Hint: `scipy.sparse.issparse`.
- `s.nnz` gives the number of non-zero elements in A .
- `s.sparsity` gives the percentage of zero elements in A .

Here, you need to import

```
from matstrucpy import matstruc
from numpy import array
from laplace0 import laplace0
```

Test your function with `A = laplace0()`.